

Mobile data services

C J Fenton, W Johnston and J D Gilliland

This paper examines the state-of-the-art mobile data services with reference to two key technologies — global system for mobile communications (GSM) and digitally enhanced cordless telecommunications (DECT).

The technology behind these two major telecommunications systems that will be providing mobile data communications for at least the next ten years is described to give a clear understanding of how the systems operate and interwork with other networks.

Proposed developments are discussed which will take the network services forward and extend them, providing the necessary capabilities into the next millennium to generate and increase the return made from the investment in building a national mobile network.

Finally, an example is given of an applications suite that provides the user with control over many services through an interactive windows environment.

1. Introduction

Data services form the foundation for the next area of market penetration and customer base expansion. Until now they have tended to be ignored in terms of a definite capability to be supported by any mobile system, and have come as an afterthought, rarely performing to an acceptable level. With the development of new networks based upon digital communications techniques, data has fitted more naturally and now provides a predictable level of service allowing applications and solutions to be planned; this development has not been easy, however, because of the need to provide backwards compatibility into analogue-based PSTN fixed networks using modems.

Developments within GSM created support for a new feature, the short message service, usually termed a value added service. This combines the capability of guaranteed data paging with datagram delivery to make a highly sophisticated messaging and notification feature.

2. General access mode

The key to successful data services is easy communication with a simple user interface. Speech is unique in that an intelligent application — the human user — can modify the interaction with the equipment so that almost

any protocol can be used to establish, modify and clear down calls. Data does not have this level of intelligence at the interface.

Communications via data requires many modules within a system, each carrying out a different and often independent process from the previous or next; these modules must interwork correctly and this is achieved by defining the boundaries or interfaces between the modules.

To fully define the capabilities of a service to be provided, the level of interaction needs to be known; for nearly all data services two categories (see Fig 1) can be defined:

- bearer services,
- telematic services.

2.1 Bearer services

These are telecommunications services providing the capability of transmission of signals between access points (called user/network interfaces in ISDN), for example, asynchronous, 9600 bit/s, 8-bit word length, 1-stop bit, out-band flow control — the interface is V.24, 25-pin.

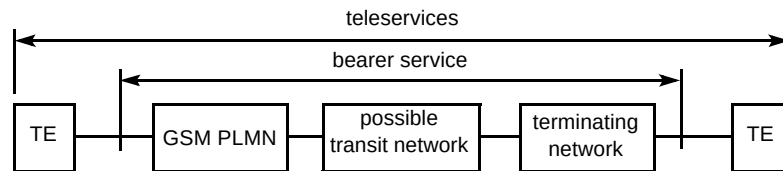


Fig 1 Categorisation of data services.

2.2 Telematic services

These are telecommunications services providing the complete capability, including terminal equipment functions, for communication between users according to protocols established by agreement between network operators, for example, facsimile group 3 — the interface is the key pad on the facsimile machine, with a place to feed in the paper copy.

3. GSM data services

GSM is a digital communications system that provides consistently reliable communications. In data communications, as the major factor that differentiates the service from digitised speech is that every bit of user data is significant, the quality of the resultant communications may vary quite widely.

The major problem area in any mobile system is the wireless communications link. For GSM a time division multiple access (TDMA)-based radio solution was chosen — this means that there are several actual communications slots within a single radio frequency channel. Each slot can be labelled differently and used for an explicit purpose. In principle there are two types of slot, the control slot (which is further subdivided for broadcast, common-control and signalling channels), and the remaining slots on any radio frequencies which form traffic channels. The basic data rate of the information within a radio frequency is high and is split among the TDMA slots; this higher rate is used to protect the user traffic by including additional information as explained below.

Data bearer services are fully ciphered in the same way as any circuit-switched communications within GSM.

3.1 Current GSM data bearers

Data rates presently used with GSM data bearers are shown in Table 1. GSM defined the communications bearers, within the GSM domain, to provide consistent data communications at an acceptable quality level. These were defined from the outset with targeted levels of performance [1], as shown in Table 2.

Table 1 GSM data rates.

X.1 rates	Sync/Async
300 bit/s	Async
1200 bit/s	Sync and Async
1200/75 bit/s	Async
2400 bit/s	Sync and Async
4800 bit/s	Sync and Async
9600 bit/s	Sync and Async

Table 2 GSM 02.08 — quality of service figures.

Bearer type	Full rate channel
Non-Transparent Async	BERmax 10^{-8}
Transparent 9.6 kbit/s	BERmax 10^{-3}
Transparent 4.8 kbit/s	BERmax 10^{-5}
Transparent 2.4 kbit/s and below	BERmax 10^{-6}

Two basic service delivery mechanisms were created with complementary performance capabilities called:

- transparent,
- non-transparent.

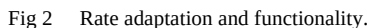
These two services can be described in services terms from the user point of view as shown in Table 3, with fuller descriptions in the following section.

Table 3 Summary of performance capabilities.

Transparent	Non-transparent
Fixed throughput	Variable throughput
Fixed delay	Variable delay
Variable error rate	Fixed very low error rate

3.1.1 Data communications model specific

ETSI-SMG defined a number of models for data interworking. The basic structure consists of terminal equipment (TE) connected via adapters to mobile terminals (MT); this concept follows the ISDN model and enables

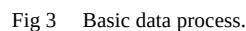


The specific mechanisms to support data services are contained within the interworking functions (IWF) unit alongside a GSM network switch. It is like providing specific transcoding capability for data which can be selected on a per-call basis by use of ISDN-like signalling. Figure 2 shows rate adaptation and modem functionality — these are within the IWF on the network side and within the mobile station or terminal adapter (described later). The IWF always resides within the visited MSC.

The data is conveyed through the system using an adapted ISDN-based data protocol, the V.110 rate adaptation scheme. The V.110 frames are modified and encapsulated for communications across the radio channel. This is handled as shown in Fig 3.

If the data is ISDN derived then the V.110 frame already exists. If the data is from another source then the first process at the GSM boundary is to put the data into the V.110 format (rate adaptation scheme RA0 [2]).

These intermediate rates allow vital control information associated with the user data and contained in the original V.110 frame to be conveyed across the radio path. From this point on the V.110 data is treated as one block of data and, whether the data be user information or GSM information, the system treats it as data that needs conveyance.



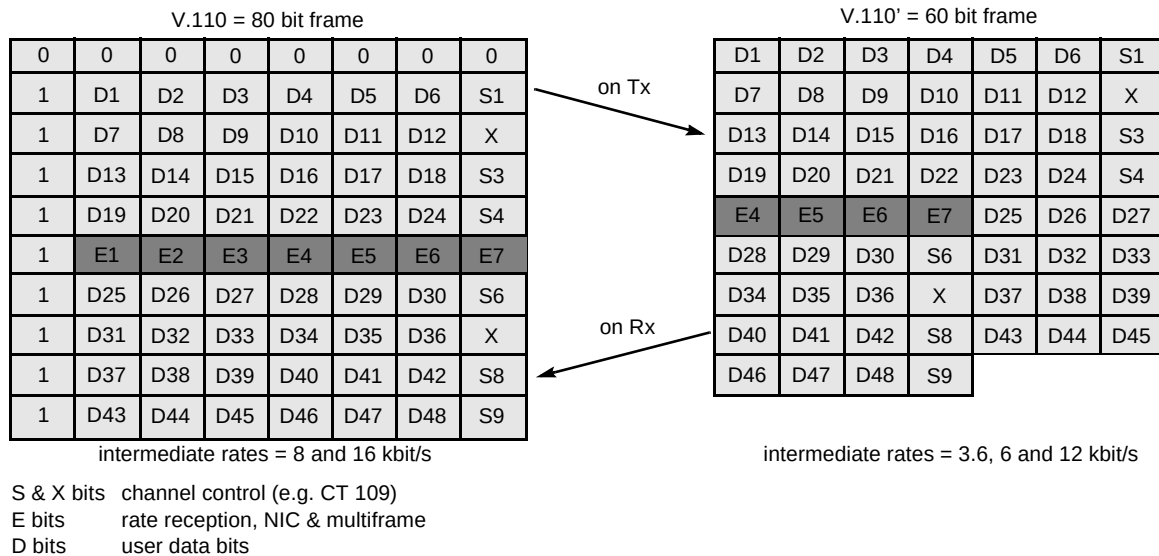


Fig 4 RA1 to RA1' function and conversion.

The radio interface provides a raw-data channel of 22.8 kbit/s and the difference between the GSM intermediate rate and 22.8 kbit/s is used for forward error correction (FEC).

While this discussion revolves around the use of an intermediate rate, the actual radio channel is segmented into short bursts. To increase the performance of the communications the modified V.110 frames are passed to the radio channel access process in a way that aligns with the radio channel TDMA bursts. With this alignment the V.110 frame can be recovered directly from the radio channel, so the standard V.110 frame synchronisation pattern can be discarded while within the radio path; this reduces the bit count from 80-bit V.110 frames to 60-bit V.110' frames allowing more FEC to be added per user data.

3.1.3 Interleaving

The effectiveness of the FEC (to correct corruption) is increased by the use of interleaving on the air interface. Interleaving separates adjacent data bits before transmission on the radio path, which means that, when data is corrupted by a radio channel burst, the errors, when de-interleaved, are separated and spread a long way apart in time. This reduces the number of errors per FEC block and enables the FEC to correct the data. Without the interleaving the corruption would appear together and reduce the ability of the FEC to correct it.

When both processes are available to system designers, a point where interleaving depth and FEC power produce the best trade-off for user data without additional delay is chosen.

The depth of interleaving was chosen to be longer than the typical fast fade on the radio channel. For GSM data services, the interleaving depth is 19 and the assignment of data to radio channel bursts is shown in Fig 5.

The delay introduced by interleaving of this type is more than might at first appear. For the first 456 bits of coded data, 4 sub-block each of 114 bits are produced; this data is interleaved over 19 burst frames but does not completely fill the burst frames. In order to fill the frames, data from the next three sub-blocks is required before the whole of the original coded data can be recovered. This means that the effective delay is that of 22×114 burst blocks:

$$\begin{aligned} \text{interleaving delay: } & 19 \times 5 = 95 \text{ ms, plus} \\ & 3 \times 5 = 15 \text{ ms} \\ \hline \text{total} & = 110 \text{ ms [3]} \end{aligned}$$

Add to this the remaining processes and propagation times, and the single passage delay is in the order of 180 ms. This is in line with that of a single hop satellite system, allowing communications protocols that communicate over satellite to continue to operate over GSM data services. The levels of FEC are given in Table 4.

Table 4 FEC levels.

User rate	Intermediate rate	FEC	Correction (for guidance)
9.6 kbit/s	12 kbit/s	10.8 kbit	48 bits in 240
4.8 kbit/s	6 kbit/s	16.8 kbit	96 bits in 240
2.4 kbit/s and below	3.6 kbit/s	19.2 kbit	160 bits in 240

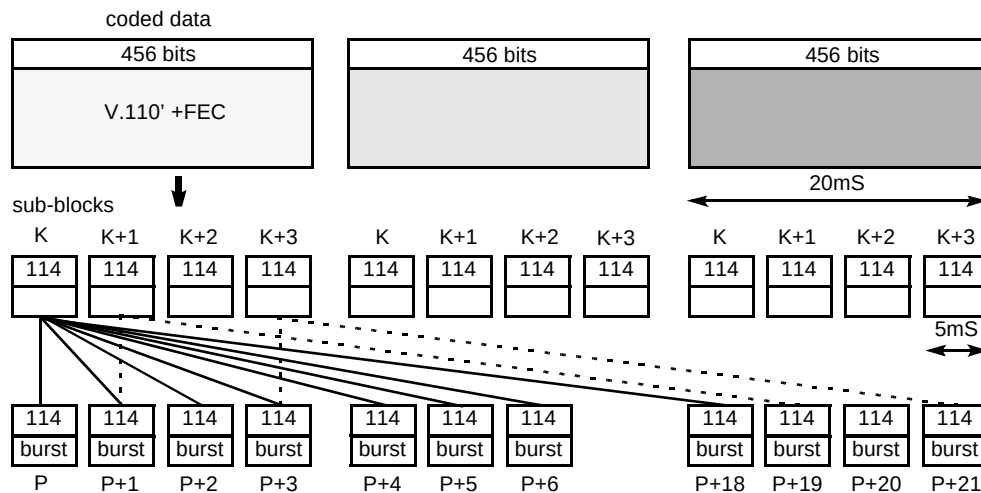


Fig 5 Interleaving on the data channel (TCH 9.6).

The whole process is shown in graphical form in Fig 6. This is the basic underlying mechanism for data services and the whole of the transparent bearer is shown. From this it can be seen that the user service profile of fixed throughput and fixed delay are met. However, should the radio-channel corruption become too great for the FEC to correct, errors will be seen in the user data. Regardless of this, GSM provides this mechanism for user applications that require these performance parameters, typically applications that support their own retransmission algorithms.

automatic retransmission, presenting error-free data to the user. This extra retransmission functionality resides both in the mobile station and the GSM fixed network interworking function (IWF) unit, considered as part of an MSC.

The retransmission process is known as the GSM radio link protocol [4] (RLP) and is optimised to interwork with the radio channel for highest performance. RLP is similar to X.25 layer 2 (Lap-B) where data units are given sequence numbers that allow either selective retransmission or go-back-N for errored block recovery. These data units are only 240 bits long with a sequence number window of 62. This allows radio-channel alignment to ease synchronisation as well as reducing the overall delay caused by errored units, since retransmission is typically of only a single 240-bit data unit. Were the length of a data unit the same as that used in the fixed network, the delays caused while a 1024-octet data unit was retransmitted until received error free would be too long and unacceptable.

When data units are corrupted on the radio link, and the level of corruption is too great for the interleaving and FEC to correct, then the retransmission starts. While retransmission is taking place user data is still being received and buffered by the GSM sending entity. Should the errors on the radio channel persist then the buffer may fill; to control this, the GSM sending entity has to be able to communicate flow control back to the originating terminal to prevent loss of data.

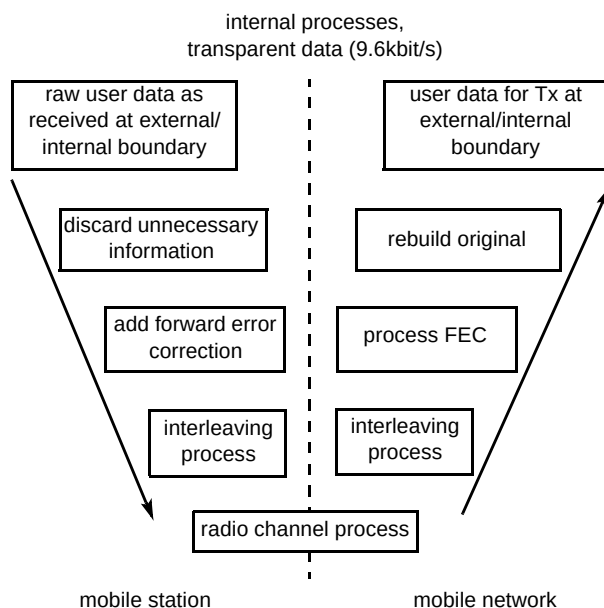


Fig 6 The interleaving process.

3.1.4 Non-transparent data mechanism

The other data mechanism is the non-transparent data service; this uses the above process and adds GSM retransmission to produce a service that has FEC plus

GSM defines non-transparent data communications for many different layer 2 protocols so that selection can be made to produce compatible flow control. This is known as the GSM layer 2 relay (L2R) function and is located between the user terminal and the radio link protocol. Various L2Rs can be selected, typically for Lap-B (X.25), ISO6429 (e.g. Xon,Xoff), and Teletex (X.75).

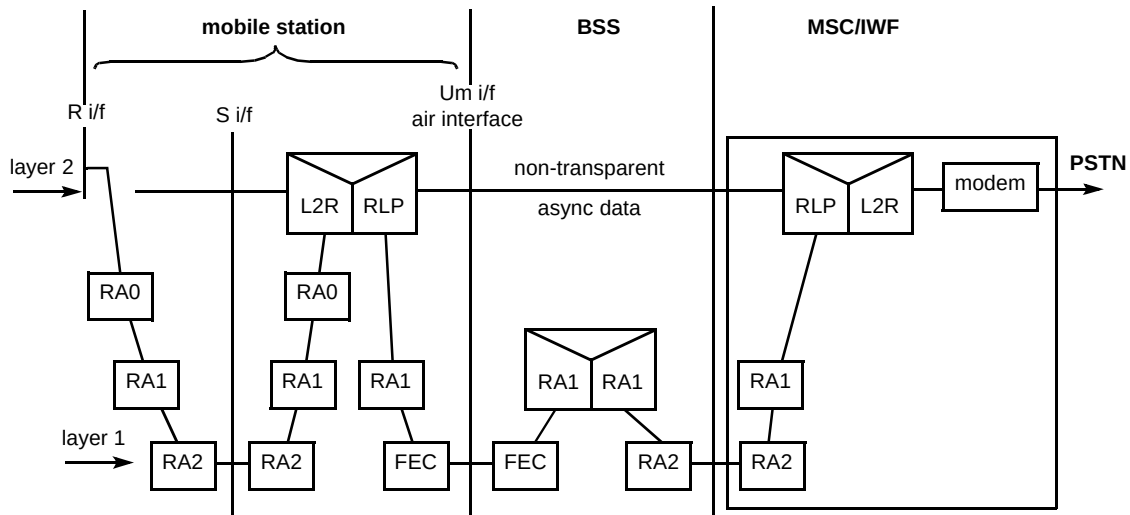


Fig 7 Non-transparent data rate adaptation.

It is possible to select out-band flow control which is matched with V.42 (error correction) in the interworking modem for the PSTN case (see Fig 7).

3.2 The data mobile

Further models and reference points for interconnection are defined for the mobile (see Fig 8).

reference points: S(ISDN)
R(non-ISDN e.g. V.24)

MT - mobile terminal
TA - terminal adapter
TE - terminal equipment

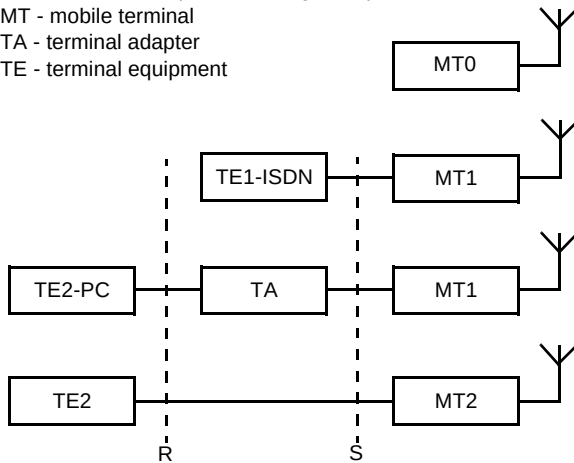


Fig 8 Mobile interconnections.

Figure 8 shows the main permutations from the MT0 class of mobile where the whole product is contained within one box and the only interfaces are the radio and the point of human intervention, e.g. a mobile facsimile machine, through to the MT2 — a mobile with a non-ISDN interface (typically V.24) providing access for simple asynchronous terminals.

In reality, the terminals that have been seen in the market-place are mainly PCMCIA (Personal Computer

Memory Card International Association, soon to be called PC Card).

In the case shown in Fig 9, the mobile conforms to the MT0 case where there are no recognised interfaces. This means that it is not possible to buy equipment from many different vendors and connect them together. However, it has led to compact solutions for data services and, with the current developments in computer communications, there will also be products that do away with the cable, replacing it by an infra-red link.

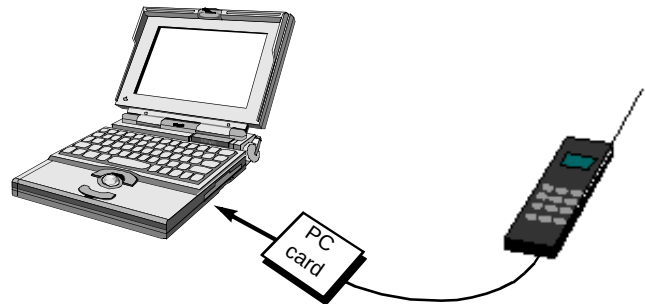


Fig 9 MT0 mobile with infra-red link.

3.3 Facsimile group 3

Facsimile is supported by use of the basic data bearer services. Its interworking with GSM is specified since normal end-to-end transmission of facsimile cannot be successful without special intervention. Facsimile is supported as shown in Fig 10.

Imagine a normal PSTN facsimile machine with a standard telephone line jack. In concept, this would interconnect to the GSM mobile by plugging into a device known as a facsimile adapter (FA) (in practice, most implementations of GSM facsimile remove the analogue portion). The FA converts the analogue modem

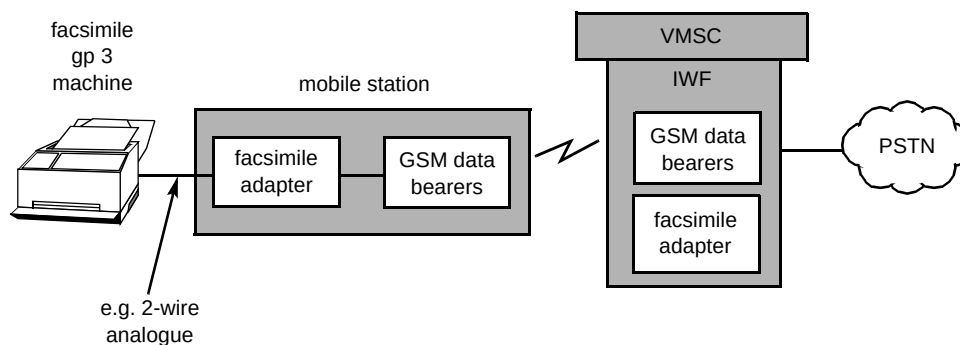


Fig 10 GSM support for facsimile.

communications back to the original digital codes and then utilises a standard GSM data bearer. At the other end of the radio link, in the corresponding device, the digital information is converted back into the analogue form for transmission across the PSTN.

The current working form of GSM facsimile utilises the transparent bearer services since these provide a fixed delay across the GSM link. This is fundamental to facsimile group 3 since it already has a layer 2 protocol within the ITU T.30 facsimile recommendation controlling the transmission of ITU T.4 coded facsimile pages. Transparent data provides the 'fixed delay/fixed throughput' service required by the facsimile protocol of a 'lower layer'.

There are many other details that could be discussed to specify facsimile fully, but space prohibits introduction here. It is sufficient at this stage to show that facsimile utilises the basic bearers of GSM; this same model is used for all other teleservices and user applications.

3.4 Interworking scenarios

GSM is designed to be for public access as a fully featured public domain communications system. To ensure that users get the most from a system of this kind, interworking to existing and emerging communications systems needs to be put in place. ETSI-SMG specified preferred interworking solutions and designed a number of solutions for consistent access so as to ease the expected interworking whenever users are roaming.

3.4.1 ISDN

GSM has many of the characteristics of ISDN in that it is digital and has common channel (or out-band) signalling.

These two similarities mean that if you stand back from GSM and look at the end points it, in fact, looks like ISDN. There are restricted service levels, but in principle GSM is an ISDN with a radio link providing the communications path; this is backed up by the use of bearer capability

information elements (including high-layer and low-layer compatibility information elements common between the two).

GSM is designed to interwork fully with ISDNs for all categories of service types. These are, in principle:

- Category 1 — 64 kbit/s PCM fully digital path required (e.g. no-speech processing allowed) — typically for data calls,
- Category 2 — A-law 64 kbit/s preferred digital or analogue routing, speech processing allowed — typically for speech,
- 3.1 kHz — 64 kbit/s path preferred, avoid speech processing equipment — typically analogue access for modems/facsimile.

For mobile terminated calls the common-channel signalling is used to deduce the GSM service that should be configured. As a basic premise, GSM requires users to have a subscription to the services they wish to use; this is managed by the network operator and stored in the network home location register against the user's international mobile subscriber identity (IMSI). Alongside the IMSI is a directory number (e.g. E.164 in their ISDN case) and this is the mobile subscriber ISDN number (MSISDN). Upon receipt of an initial address message (IAM) from the national fixed network the GSM interrogating node (IN) converts the IAM information into GSM equivalent values (described in GSM TS 09.07) (see Fig 11).

From these values a GSM service can be deduced and this can be validated against the subscriber record in the HLR to check subscription. Assuming this is validated, the normal mobility management process takes place and an attempt is made to set up the desired call to the mobile.

For ISDN digital working, rate adaption rather than modems are used. The adapters are held in the IWF associated with the VMSC in which the mobile is currently located. This principle also applies if a mobile is roaming to another network where the IWF of that network is used.

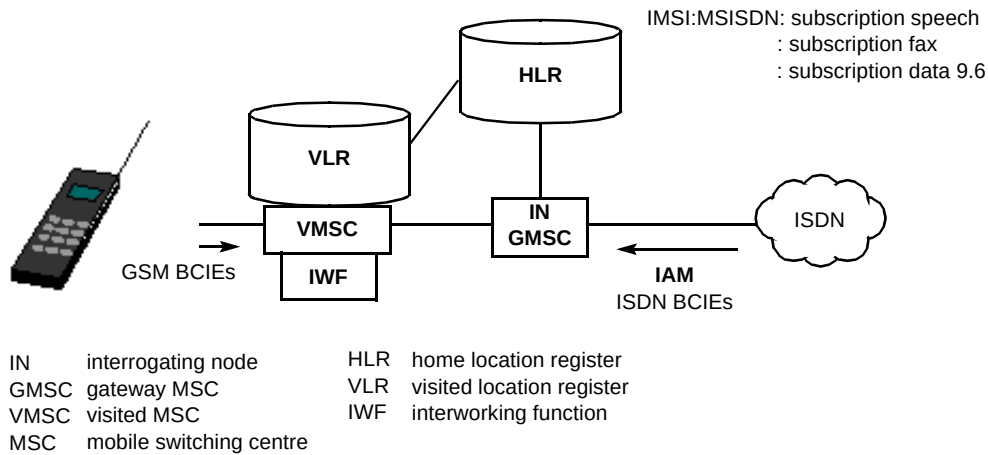


Fig 11 GSM/ISDN interworking.

In the case of ISDN interworking there are a number of GSM-specific BCIEs that are not supported within the national ISDN signalling, for example transparent or non-transparent. A mobile is offered a data call from the basic group that is derived from the IAM. The offering from the GSM PLMN contains some BCIEs that can be negotiated by the mobile, T or NT is a good example. A GSM network may support T, NT or both, this is the case for most known networks. As a result the offering to the mobile is made such that there are no restrictions. Most mobile networks offer 'both with NT preferred' in the set-up signalling. This does not restrict any mobile that could be less capable and only support, for example, T.

Mobile-originated calls work in the opposite sense, in that GSM values are mapped into the ISDN scheme and calls continue in the normal way.

3.4.2 PSTN

Interworking to the PSTN poses a major problem since there are not the sophisticated facilities of common channel signalling. As a result, the PSTN caller cannot indicate the

GSM service they require to invoke. To overcome this limitation, use is made of the so-called 'multinumbering system'.

In this, a mobile subscriber has an IMSI, as introduced above, and associated with the IMSI is a set of services to which they subscribe. In addition to the list of subscriptions, there is a matching set, one for one, of diallable MSISDNs (see Fig 12).

The PSTN user can now identify the service that is required of the GSM network by dialling the specific number relating to the service desired. The GSM HLR produces the appropriate BCIEs which are forwarded to the VMSC and service continues as for the ISDN case.

Mobile-originated calls pose no problem for interworking (only restrictions to signalling-based information, e.g. calling line identity) to the PSTN since, at the network's interconnect point, the GSM data is discarded as it has no equivalent in the PSTN to which it can be mapped.

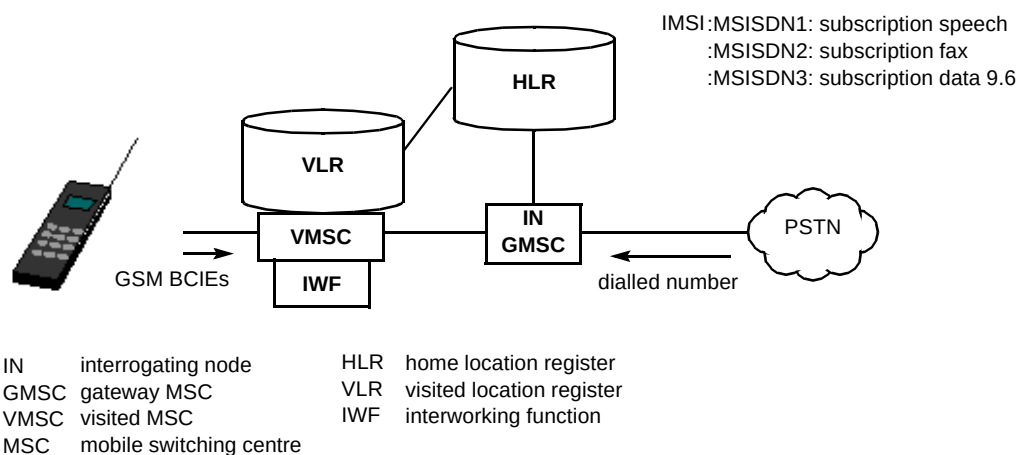


Fig 12 GSM/PSTN interworking.

4. GSM short message service

The GSM teleservice, short message service (SMS), is an example of where GSM managed to build on the user needs by including a messaging mechanism; this is one of the first and now most widely used 'value add' services that exists anywhere in any telecommunications system. Significantly, SMS mobile terminated is the only mandatory service that mobile stations have to support.

There are two services within the SMS portfolio:

- SMS point to point — mobile terminated and mobile originated,
- SMS cell broadcast.

As SMS services are supported on the signalling channels which are almost constantly handling communications between the mobile terminal and the network, transmission does not require a traffic channel and can take place alongside circuit-switched communications.

4.1 SMS point-to-point

Point-to-Point messages are communicated between a mobile and the SMS service centre (SMS-SC). This SMS-SC is considered to be outside the GSM PLMN and is managed by an operator who is usually the same as the GSM PLMN operator. As this SMS-SC resides outside the GSM PLMN it is also outside the scope of any GSM specifications, and interworking between the GSM PLMN and the service centre is specific to each network operator (see Fig 13). The SMS-SC connects to the GSM PLMN normally by use of Signalling System No 7, but other options are not prohibited.

SMS PTP message communication is limited to one message, that is to say there is no relationship within GSM between SMS messages. These messages have a user information field of 140 octets, normally this relates to 160

characters, but the information field is open to be used as required by the implementor of the application.

SMS PTP messages are always relayed through an SMS-SC. As the SMS-SC resides outside the GSM network and is connected by signalling links, mobile-originated messages can communicate with any service centre via the national/international signalling system.

There are many uses of SMS PTP, for example:

- notification of delivery of facsimile, e-mail, voice messages,
- short e-mail, e.g. mobile-to-mobile text,
- database enquiry.

In order to make these services complete one of the fundamental features of SMS PTP is that it has a close relationship with the availability of a mobile. When the SMS-SC has a message for the mobile it attempts to delivery it by communicating with the GMSC and subsequently the HLR. The mobile is identified by its international directory number (MSISDN — mobile station international ISDN number).

If the mobile is available then the SMS message progresses by following the normal interrogation procedure as used in the call control procedure; this invokes all the subscription, and encryption procedures as for a normal mobile-terminated switched-circuit call. It should be noted that for each SMS PTP message there is a considerable overhead and SMS PTP is not an alternative for packet-oriented data services.

If the mobile is unavailable then markers are set in the V/HLR (depending upon the sort of unavailability — e.g. not reachable (VLR) or detached (HLR)). These markers also enable the PLMN to identify the SMS-SC that originated the message. When the mobile becomes available again, a message is sent by the HLR to alert the relevant SMS-SC to retry delivery of the stored messages.

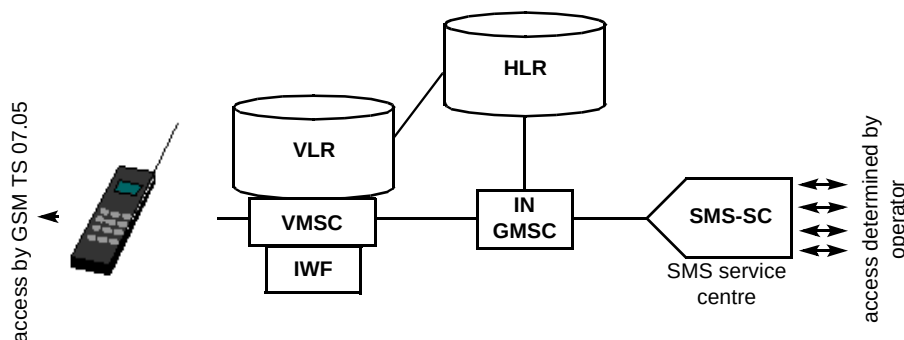


Fig 13 GSM/SMS-SC interworking.

SMS has many features built in to make it a sophisticated messaging system — two such features are:

- replace short message — enables a subsequent message to overwrite the relevant previous message, thereby optimising the utilisation of phone or SIM memory, news or notification being updated without being rejected by the mobile because of a full memory,
- numbers in quotes (“123”) — when reading an SMS message on the mobile screen, any numbers in quotation marks can be selected and used to dial a mobile-originated call, selection being achieved by pressing the ‘send’ button on the mobile and the out-dialling starts.

4.2 SMS cell broadcast

Cell broadcast is a mechanism that enables a network operator to transmit information that is distinct within a cell. Each cell has its own identity and can contain completely different cell broadcast information (see Fig 14).

A cell broadcast service centre (CBSC) maintains the distinct information for broadcast within each separate cell. Access to the CBSC is operator determined.

A mobile terminal equipped for SMS-CB can receive information while in the idle mode, i.e. while not engaged in a circuit-switched call. Information is transmitted in each cell with a reference number. Transmissions are repeated in a constant cycle, the concept being rather like the television teletext process where the display is selected by nomination of the required reference number on the mobile. As that page is transmitted, it is displayed on the mobile station. Each page contains 82 octets (normally taken to be 93 characters).

Examples of the the sort of information for transmission are:

- network information,
- weather, travel (road problems),
- local information:
 - hotels,
 - garages,
 - events,
 - promotions,
- zone trafficking (Cellnet Call Saver).

Information can change as regularly as the operator of the broadcast centre allows.

5. DECT data service

The digitally enhanced cordless telecommunications (DECT) standard is being developed by ETSI so that both voice and data traffic can be delivered over the system. Six data profiles have been identified that cater for most of the present, and some future, data delivering for up to twelve simultaneous telephony calls. DECT can also be configured to transport an error-corrected 525-kbit/s throughput using 23 of the 24 time-slots of transmission in one direction. During transmission the capacity can be dynamically varied.

The interoperability of equipment from different manufacturers is of prime importance for the success and cost reduction of mobile data services and ETSI is working towards common standards in this area.

5.1 Applications

DECT will need to be able to support many existing applications as well as applications particular to wireless connections. Some of these are:

- group 3 and group 4 facsimile,
- e-mail,

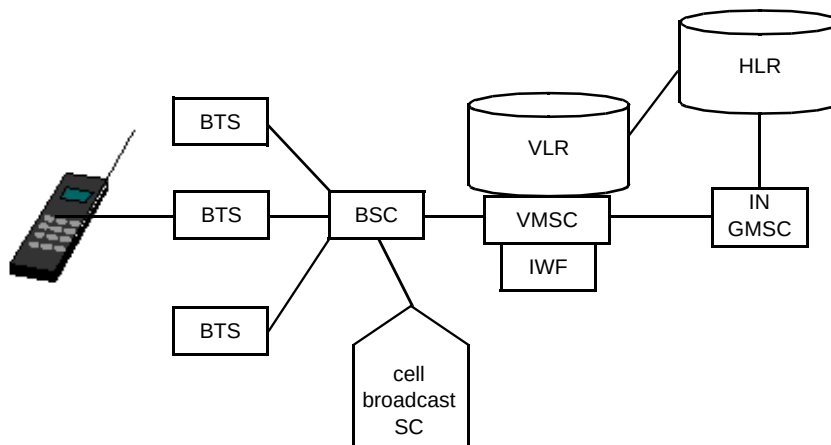


Fig 14 The cell broadcast mechanism.

- real-time video,
- file transfer,
- server access,
- multimedia,
- printing,
- paging.

Each application has its own special requirement for data transport, some requiring high data rates and others, for example video, requiring time-bounded data transport, i.e. low delay. It is important that the implications of these requirements are well understood so that the wireless connection can meet the relevant requirements. Figure 15 shows some services with their data rate and delay requirements.

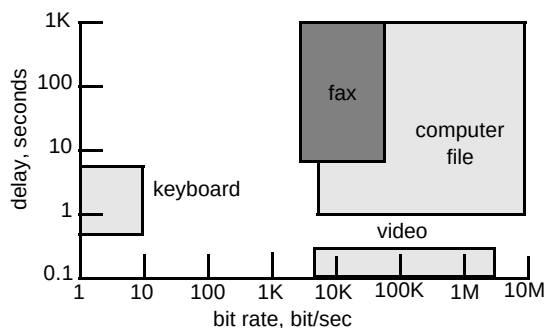


Fig 15 Data services.

In order to offer some of these services an operator would need to ensure that the wireless connection can meet the bit error rate performance for a particular service. The services which can tolerate long delays can tolerate short bursts of high bit error rate on the wireless link, since errored frames can be retransmitted. Services such as video telephony, that cannot tolerate long delays, need a better grade of wireless link, since errored frames will not be retransmitted; instead they will be discarded and there will be a corresponding degradation in the service quality.

5.2 DECT frame structure

DECT uses a time division duplex (TDD) frame structure of twelve transmit and twelve receive time-slots in one frame of 10 ms duration as shown in Fig 16. This can support twelve simultaneous data duplex channels of 24 kbit/s.

Whenever a higher data rate is required the time-slots can be concatenated. The maximum capacity that can be provided is when asymmetrical use is made of the DECT frame by concatenating 23 time-slots for transmission in one direction, leaving 1 time-slot for transmission in the

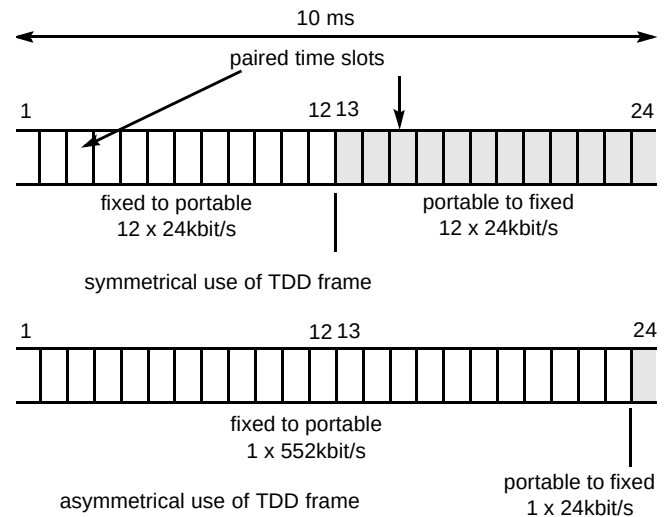


Fig 16 DECT frame structure.

reverse direction. In this way the maximum data rate of 552 kbit/s can be asymmetrically transmitted.

5.3 Network

The above applications should be able to work over the following networks:

- PSTN, ISDN,
- LAN, MAN, WAN,
- GSM.

5.4 Mobility

Mobility for terminals will be a very important part of the data system and DECT is able to support two types of mobility:

- local area mobility,
- roaming between networks.

Figure 17 shows a scenario where a multicell coverage of a building gives both handsets and portable computers local area mobility within the area covered by the cells. The intelligence for handover between cells resides within the DECT controller.

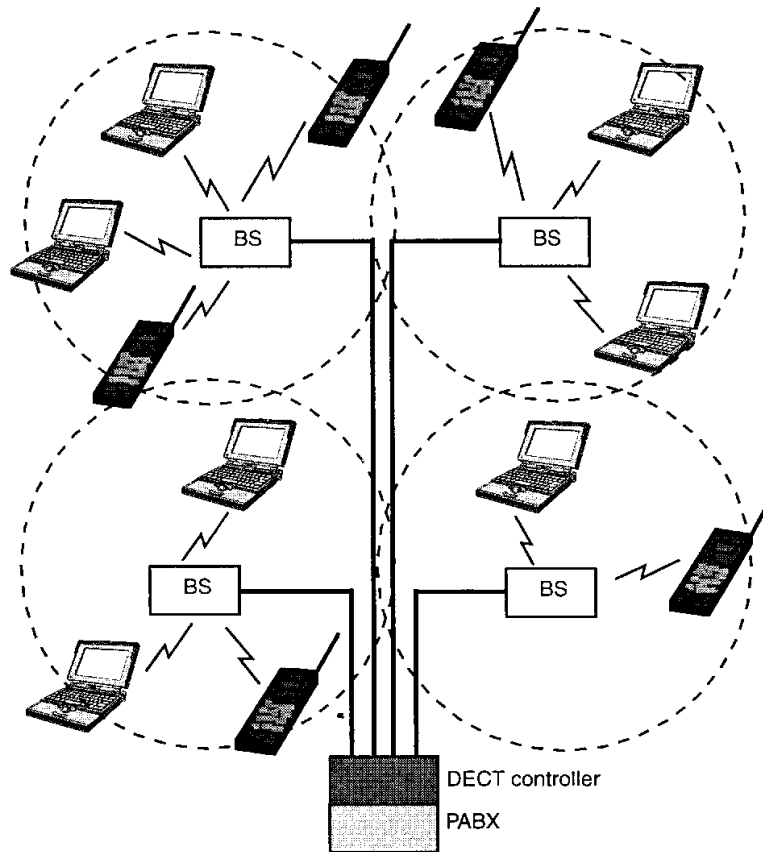


Fig 17 Cellular cordless LAN for local area mobility.

Figure 18 shows a scenario where DECT portable equipment can operate in two separate locations with the mobility intelligence residing in the network.

5.5 Interoperability and security

Part of the approach to ensure low-cost portable part (PP) and fixed part (FP) DECT equipment is to have a standardised air interface between the PP and FP so that equipment from different manufacturers can interwork (as shown in Fig 20 in section 5.7). ETSI is working to ensure that the DECT data profiles meet the interoperability requirement.

The generic access profile (GAP) for voice services has encryption to provide security over the air interface. The mobility provided for roaming between networks embodies the GAP encryption, thereby ensuring a very secure data channel for both private and public data services.

5.6 Service types

ETSI have defined six DECT data service profiles [5] in order to meet the wide variety of applications, and these are shown in Table 5.

Table 5 Service type description.

Service type	Description
Type A	Low-speed frame relay with up to 24 kbit/s throughput. Optimised for bursty data, low power consumption and hand-portable equipment.
Type B	High-performance frame relay data with 422 kbit/s throughput optimised for high speed and low latency with bursty data. Equipment implementing type B profile shall interoperate with type A equipment.
Type C	Non-transparent connection of data streams requiring link access protocol (LAP) services, optimised for high reliability and low additional complexity. This builds on the services offered by type A and B profiles. Provision for a packet assembly and disassembly function for asynchronous data streams is also included.
Type D	Transparent and isochronous connection of synchronous data streams optimised for interworking applications requiring continuous data streams.
Type E	A short message transfer or paging service which may be unacknowledged or acknowledged, optimised for small service data units, low PP complexity and ultra-low power consumption.
Type F	An application profile specifically supporting teleservices, such as facsimile, building upon the services offered by the type A, B and C profiles, optimised for terminal simplicity, spectrum efficiency and network flexibility.

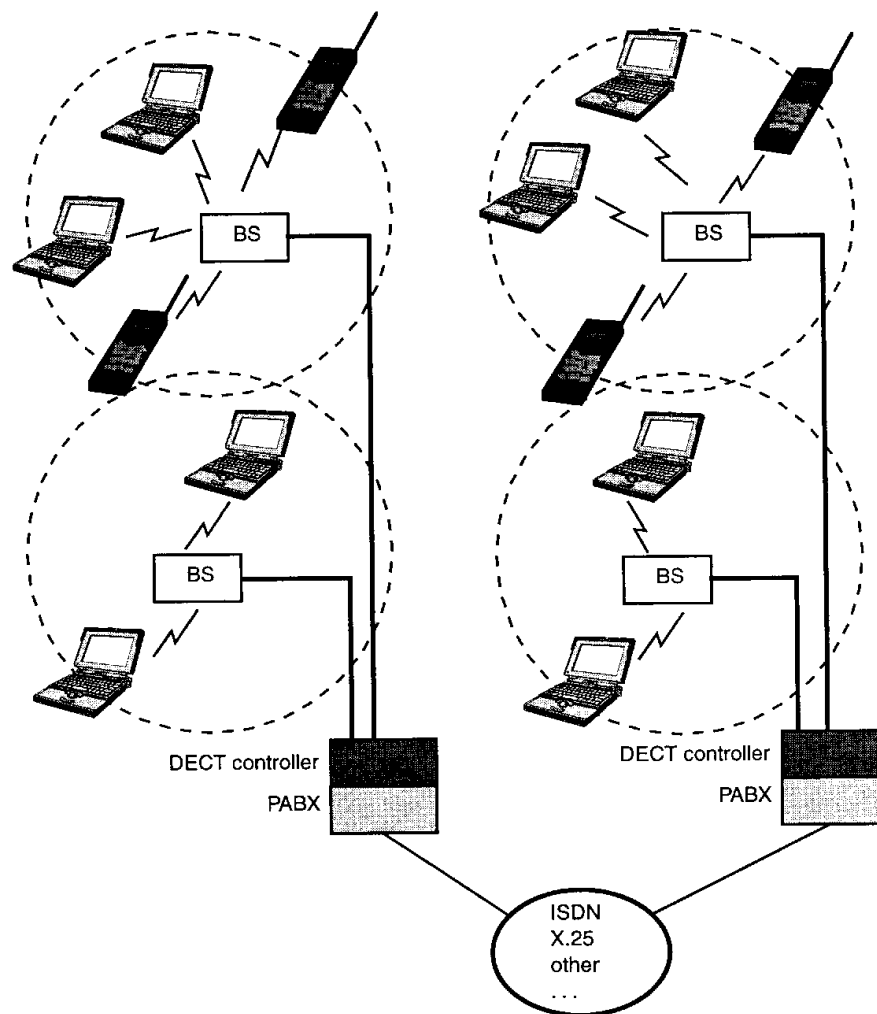


Fig 18 Roaming between networks.

Figure 19 shows the relationships of the six service types.

5.7 Profiles

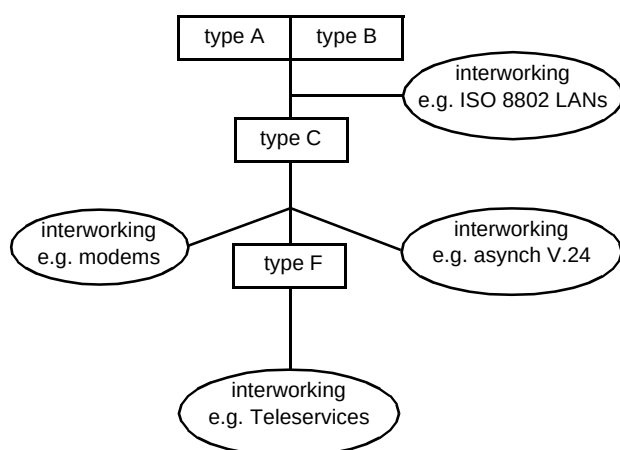


Fig 19 Service type relationships.

Each of the six profiles have been designed for particular sets of services, applications or type of equipment. A fuller description is given below.

- **Type A profile**, with local area mobility capabilities, uses a single duplex DECT bearer which ensures low power drain and this makes it suitable for hand-portable equipment. This profile was adopted in December 1995.
- **Type B profile**, with local area mobility capabilities, uses the multi-bearer provision in DECT, and may also use the asymmetric-channel feature for maximum uni-directional data transfer. This profile is suited to multimedia applications. This profile, with local area capabilities, was adopted in December 1995.
- **Type C profile**, provides flow control and extra error recovery. It also has mobility management functions to support roaming in both private and public

environments. This profile supports interworking with V.24 circuits and is ideal for wireless connection to printers and modems using RS-232 links with data rates up to 240 kbit/s. This profile, with roaming between networks capability, should be adopted in the summer of 1996.

- **Type D profile** is still under discussion, but will be designed for security applications and radio local loop (RLL) services. It is expected to be completed in 1997.
- **Type E profile** is used for low-rate messaging service (LRMS) for point-to-point or point-to-multipoint (PTM) applications. Using the unacknowledged point-to-point low-rate message service, a network server can control a subgroup of terminals with single multicast messages without the need for a full bi-directional link. The LRMS PTM may co-exist with other profiles in the PPs and FPs and should not have any effect on the functionality of the co-existing profiles; the roaming between networks mobility part of this profile should be adopted at the end of 1996.
- **Type F profile** uses the multimedia message services (MMS) [6] which is a set of commands for file transfer and messaging. It can be regarded as a DECT internal teleservice that can be interworked to similar services

in external networks. This messaging service allows interworking to other message-based teleservices and application-level services, such as:

- e-mail,
- World Wide Web (WWW),
- hypertext transfer protocol (HTTP),
- file transfer (FTP, FTAM and ISDN file transfer).

Figure 20 shows the way MMS connects to other services via the messaging interworking unit (MIWU). MMS is transported over the multimedia message services protocol (MMSP). This profile should be finished by the end of 1996.

5.8 Structure of service profiles

DECT data service profiles do not use the 'modem over the voice channel' mechanism for data transfer, but instead transfer data directly into the data transmission mechanism of the DECT system. This gives digital transmission between the user equipment and the network providing higher throughput than can be achieved with modems.

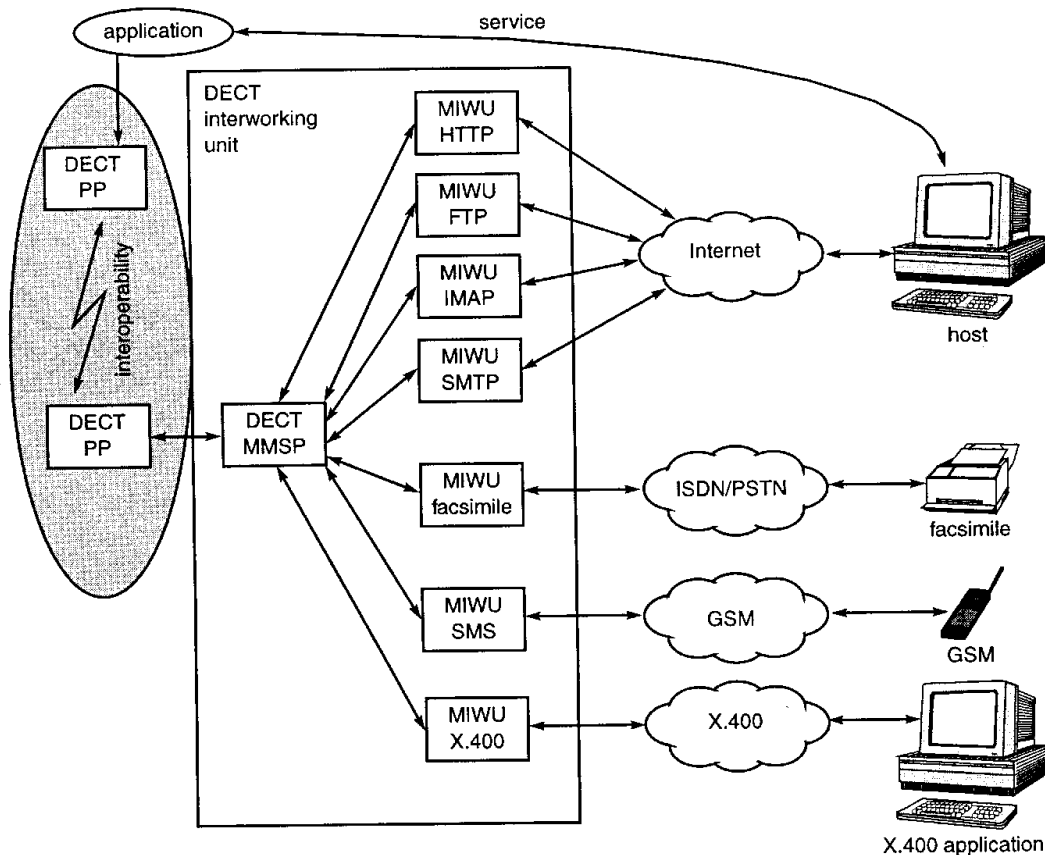


Fig 20 MMS capabilities.

ETSI are writing standards for the above data services, but not all of these have yet been published. When the standards are complete, DECT will offer a very useful and spectrum-efficient means to give mobility to an extensive array of portable equipment needing a wide range of bit rates and interconnection to other services.

6. Key comparisons

There are fundamental differences between GSM and DECT.

DECT data is based around a 24 kbit/s bearer which is used and connected to as required by the particular user application. DECT's 24 kbit/s bearers can be concatenated to provide higher data rates on a by-demand basis. These bearers are all transparent to the user data and any retransmissions **must** be managed by the user application.

GSM is almost the opposite, providing a basic single time-slot within a radio interface that has capabilities, up to a limit, for automatic retransmission, so the user does not need a complicated application to manage errors caused by communications through the radio interface.

7. Future GSM developments

GSM is undergoing continual development ranging from network enhancements to remotely locate the IWF at a central point allowing sharing of the data resource between all network switches, to proposals to modify the radio interface coding schemes to be able to support a channel rate of 14.4 and 28.8 kbit/s, developments that may not see the same light of day for some time because of their complexity. The key data developments that are likely to be seen in the market in the immediate future are given below.

- Data compression

Data compression has recently been added to the non-transparent data bearer services so that the internal transport of user data will effectively be higher than the channel symbol rate. This feature is based around the addition of V.42bis within the L2R as shown in the non-transparent data model. It is up to the network operator how this compressed data is to interwork with fixed networks where mapping of GSM data is either re-compressed in a modem link, or use of a higher speed modem or different ISDN rate adaptation scheme provides the end-to-end data rate improvement.

- General packet radio services

To date GSM has only defined and implemented circuit-switched data services. These have been designed for the fading radio environment and have proved to work well. This is the key difference

between GSM and preceding mobile analogue systems as GSM provides a level of operation that can be relied upon. However, the use of circuit-switched circuits does not make for the most efficient use of network resources particularly since most data is of an infrequent bursty nature where most of the circuit-switched connection time is not used.

GSM is developing a new data bearer which is to be packet data based; this is called the GSM general packet radio service (GPRS) and will communicate with many different data sources but only consume GSM network resource when data is actually being conveyed. It will be particularly useful when interworking with packet data networks (PDNs), especially X.25 and TCP/IP which are predominant in the market-place.

- High-speed circuit-switched data (HSCSD)

To improve the rate of communication, GSM is developing a new service which proposes to combine several traffic channels resulting in a higher data rate. The proposal also suggests that data compression should be available within the concatenated bearers increasing throughput further. There is to be concatenation of both transparent and non-transparent data bearers (however, not mixed types).

In concept, the performance will be from the current single traffic channel up to a full radio carrier or 8 time-slots. The maximum data rate being in the order of 80 kbit/s (channel symbol rate), on top of which data compression will be operating, providing two or three times the throughput depending upon the user data.

8. Applications

There are three ingredients which contribute to the success of any new service or technology:

- a common set of standards needs to be formulated and agreed to provide a foundation upon which manufacturers can develop and build the equipment that implements the new ideas,
- the equipment itself must be available for deployment and meet the standards specification,
- customers must be willing to use and pay for the system.

While it is essential for the first two requisites to be in place to ensure operator acceptance and widespread deployment there is little point to the time and investment by manufacturers and operators if the customer either cannot understand the service or see how it will benefit them. Without this, customers will not wish to purchase or use the system.

Customers generally have little interest in either the standards or the technology being used to deliver a mobile service. Their main concern is that the service works when they need it, the associated products are easy to use and the total solution makes their life easier or better in some way. Speech on mobile phones has become accepted by the general public following years of exposure to the cellular technology, so it is not difficult to see or sell the benefits of an improved system which offers security and greater clarity of speech. Data services are not, however, in widespread use and it requires more effort to explain the benefits and uses. One way to explain the benefits of reliable data bearer services is to show and market typical applications that customers already use but operating over a GSM system; this may also be aided by 'data friendly' products.

Probably one of the most widely used data services is facsimile. Most people have seen and used a facsimile machine and are aware of its potential and uses. It is therefore one of the key applications to sell initially as a mobile solution. Since the public already know the benefits of facsimile it is a small step to explain the benefits of sending and receiving a facsimile while on the move. In fact, facsimile is one of the first data services offered by many cellular operators.

BTL has developed a demonstration called 'Office On The Move' which shows how a mobile user could have all communications requirements met in one integrated package. Figure 21 shows the front screen of the demonstration. The demonstrator is an integrated suite of applications showing various aspects of data over GSM. Each application is presented as an icon and can be run from the front screen. The e-mail, facsimile and Internet applications all run standard off-the-shelf software in the way a modem-based service would operate. These applications are already popular among corporate customers and with a growing number of small business and domestic customers. They use all the available types of data service currently available on GSM and aim to encourage use of the data services.

SMS is a very GSM-specific service and is proving useful, initially, as a notification service, alerting the user to waiting voicemail, e-mails and facsimiles.

As described earlier, there are two categories of data, transparent and non-transparent. Because of the time-critical requirements of some applications it may be necessary to run the applications in one mode rather than another. One prime example is facsimile which uses the transparent data service because it is a time-critical application. The key characteristic of the transparent service

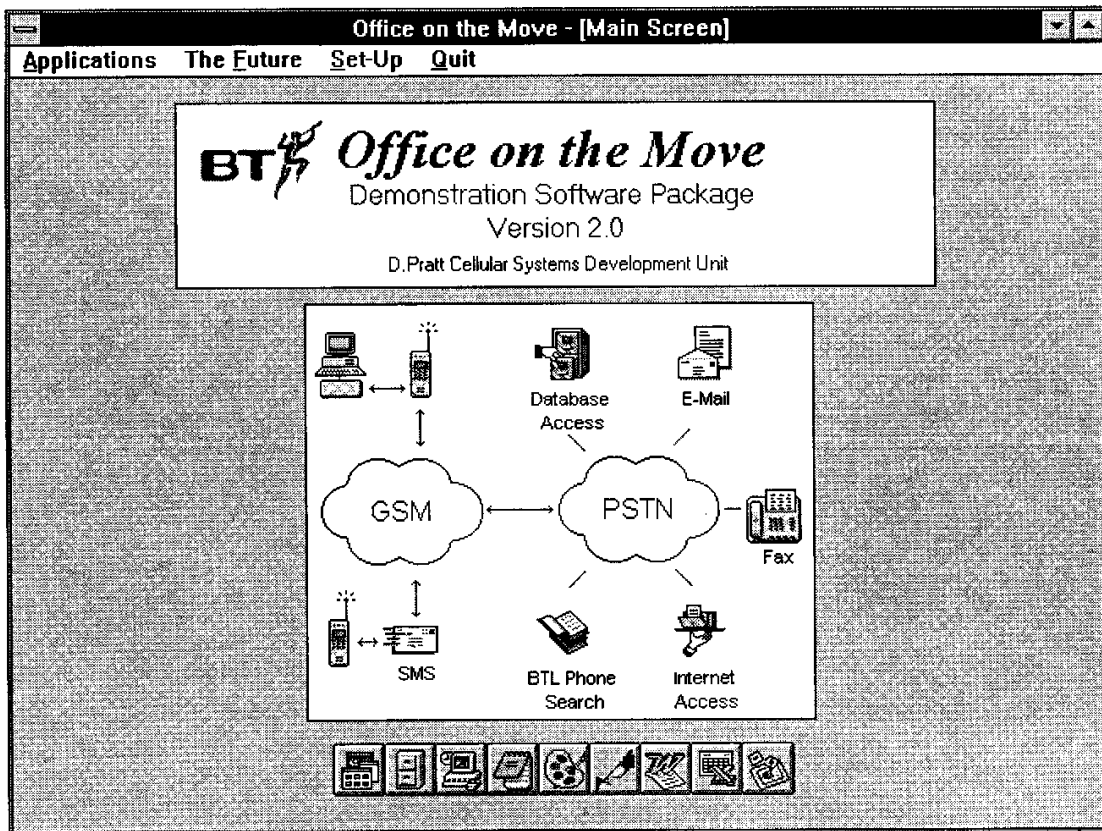


Fig 21 Office on the move front screen.

is its known fixed delay. Other applications may include real-time systems such as video.

One of the advantages of the GSM system is that, as it is backward compatible with other technologies such as standard facsimile machines and modem-based applications, there are thus immediately numerous software applications available off the shelf that can be configured and used. The configuration only generally requires changes to the modem type and configuration that would normally be required to configure the application. This situation has arisen because of the consolidated standards work prior to equipment manufacture. DECT has followed a similar life cycle starting with European-wide agreed standards. As a result DECT is also compatible with existing technologies, old and new and thus offers the same advantages of compatibility with existing applications and communications systems. In addition, the way is open to allow interconnection of DECT and GSM technologies at many different levels. Although DECT equipment is not yet in abundance it has the same advantage as GSM when it comes to potential applications.

Coupled with the need for applications to use the data services is the need for products which make it easy to use data services. To date, cellphones have been manufactured for speech applications. Products are now becoming available that are aimed at an integrated voice and data solution. Initial products are a combination of organiser and palm-top computer facilities with integrated communications capabilities allowing both speech and data. These products are the first step to providing a one-box integrated solution. The combination of a growing range of mobile specific products with a multiplicity of applications will result in simulating the market-place further and help sell data services.

As well as combining computing with communications features, other products are combining radio technologies such as GSM and DECT into one product [7]; this gives the advantage of country-wide roaming with the cheapness and privacy of a corporate PBX. Given that the two technologies both support all main communications standards from the PSTN to the ISDN and are compatible with each other, it will also allow any data applications and services to run on both systems, enhancing further the opportunities to increase the market for mobile data services.

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Chris Fenton joined BT in 1980 and subsequently attended Kent University, Canterbury where he obtained a BSc(Hons) in electronic engineering. On leaving university he joined BT Mobile in London where he worked on mobile data for vehicle location systems and BT's national paging network. In 1989 he moved to BT Laboratories and joined the cellular systems section to develop cellular data services mainly for Cellnet's GSM system. He has chaired a subgroup of ETSI SMG4 on data and telematic services for four years.



William Johnston joined BT in 1989, after obtaining a BEng (Hons) in electronic and electrical engineering at Robert Gordon's University, Aberdeen. Initially he worked in the radio division on cordless systems concentrating on CT2 and DECT. This work included data over CT2 and DECT and showed real time video over DECT.

He joined the cellular systems development unit in 1994 and is currently working on data solutions and applications over GSM.



Don Gilliland joined BT in Belfast as an apprentice in 1963. After three years at Queen's University, Belfast he obtained a BSc in electrical engineering in 1970. He then moved to Dollis Hill to work on the digital trunk microwave project. In 1980 he moved to the microwave technology group in the radio division to work on experimental microwave components. He returned to the digital trunk group in 1985 and worked on interference models for PEACEMAKER, the radio frequency assignment tool used by BT, before progressing to work on radio local loop access systems. He has attended the

ETSI standards committee for DECT over the last two years.